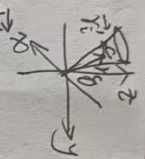


$$\vec{L} = \sum \vec{r}_i \times m_i \vec{v}_i$$

$$= \sum (\vec{r}_i + \vec{r}_c) \times m_i (\vec{v}_c + \vec{v}_i')$$

$$= \sum \vec{r}_i \times m_i \vec{v}_c + \sum \vec{r}_i \times m_i \vec{v}_i'$$

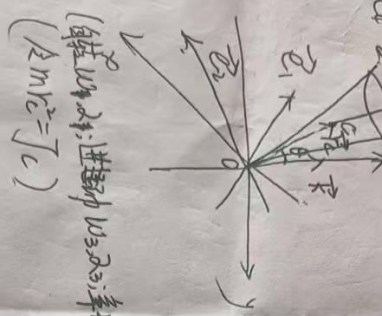
$$+ \sum m_i \vec{r}_c \times \vec{v}_c + \sum m_i \vec{r}_c \times \vec{v}_i'$$



$$\sum m_i \vec{r}_c \times \vec{v}_c = \vec{r}_c \times \sum m_i \vec{v}_c = \vec{r}_c \times M \vec{v}_c = \vec{L}_c$$

$$\sum m_i \vec{r}_i \times \vec{v}_c = \vec{r}_c \times M \vec{v}_c = \vec{L}_c$$

$$\sum m_i \vec{r}_i \times \vec{v}_i' = \vec{L}'$$



(选择质心为进动轴，角动量守恒)

($\Delta m \vec{v} = \vec{L}_c$)

$$\vec{p} = \vec{p}_c = \vec{p}_{c进} + \vec{p}_{c转}$$

$$\vec{p}_{c进} = m \vec{v}_{c进} = m \omega \vec{r}_c \sin \theta \vec{e}_2$$

$$\vec{p}_{c转} = m \omega \vec{r}_c \sin \theta \vec{e}_4$$

$$\vec{r}_c \times \vec{p} = m \omega \vec{r}_c \sin \theta (\vec{r}_c \times \vec{e}_2) + m \omega \vec{r}_c \sin \theta (\vec{r}_c \times \vec{e}_4)$$

$$= m \omega \vec{r}_c \sin \theta (\vec{e}_4 - \vec{e}_2) = m \omega \vec{r}_c \sin \theta (\vec{e}_4 - \vec{e}_2)$$

$$\vec{L} = \vec{r}_c \times \vec{p} = m \omega \vec{r}_c \sin \theta (\vec{e}_4 - \vec{e}_2)$$

$$\vec{L} = m \omega \vec{r}_c \sin \theta \vec{e}_2$$

$$\frac{d\vec{L}}{dt} = \frac{d}{dt} [m \omega \vec{r}_c \sin \theta \vec{e}_2]$$

$$= m \omega \frac{d\vec{e}_2}{dt} + \frac{d}{dt} [m \omega \vec{r}_c \sin \theta \vec{e}_2]$$

$$= m \omega \frac{d\vec{e}_2}{dt} + m \omega \vec{r}_c \sin \theta \frac{d\vec{e}_2}{dt}$$

$$= m \omega \frac{d\vec{e}_2}{dt} + m \omega \vec{r}_c \sin \theta \frac{d\vec{e}_2}{dt}$$

$$= m \omega \frac{d\vec{e}_2}{dt} + m \omega \vec{r}_c \sin \theta \frac{d\vec{e}_2}{dt}$$

$$= m \omega \frac{d\vec{e}_2}{dt} + m \omega \vec{r}_c \sin \theta \frac{d\vec{e}_2}{dt}$$

$$\int \omega_1 \sin \theta + \int \omega_2 \cos \theta - \frac{1}{2} \int \omega_3 \sin \theta - \int \omega_4 \sin \theta = 0$$

$$\int \omega_1 \sin \theta - \frac{1}{2} \int \omega_3 \sin \theta + \int \omega_2 \cos \theta = m \omega \vec{r}_c \sin \theta$$

$$\int \omega_1 \sin \theta - \int \omega_2 \cos \theta + \int \omega_3 \sin \theta + \int \omega_4 \sin \theta = 0$$